

Explainable AI for Diagnosing Acne Severity Using Smartphone Images

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Abstract—Acne vulgaris is one of the most common skin ailments, and as a result, to efficiently treat acne vulgaris, a clinician must assess the severity of the acne vulgaris. Due to geographic limitations, acne vulgaris is often approached with subjective assessments and specialist access is very limited, especially in developing countries. Therefore, to improve the objective evaluation of acne vulgaris severity through the use of smartphones, we developed an Explainable AI(XAI) model that combines an EfficientNet-B0 based deep learning model with Grad-CAM, LIME, and SHAP methods for transparency and explainability. The hybrid implementation of our framework using a clinical and smartphone dataset was tested and achieved a 91.6% accuracy across the entire dataset. The models showed correspondence between the areas of the skin that were identified by clinicians as most prominent for the severity of acne vulgaris and the areas that the model focused on when making its predictions about severity. Moreover, the framework shows consistent performance across different skin tones and imaging conditions under realistic tele-dermatology scenarios.

Index Terms—Acne Vulgaris, CNN, Grad-CAM, Medical Image Analysis, Skin Disease Classification, Smartphone-Based Diagnosis.

I. INTRODUCTION

Acne vulgaris is a common skin problem that affects the people worldwide, especially teenagers and young adults. The condition ranges from mild comedonal to severe nodulocystic forms, accurate and timely severity assessment for establishing an effective treatment planning [1]. Traditional acne diagnosis is primarily depends on visual assessment by dermatologist, which introduce subjective and dependent on clinical expertise. Furthermore, the limited availability of dermatological services in rural and underserved areas, underscores the urgent need for automated systems to provide accessible and consistent diagnostic solutions [2].

Recent advances in artificial intelligence (AI), mainly deep learning and computer vision, have shown remarkable success in medical images analysis tasks [3]. Convolutional Neural Network(CNNs) are extensively adopted for skin disease diagnostics due to their ability to automatically extract critical features from images without manual intervention [4].

The increasing availability of high quality of modern smartphone cameras for facial images for dermatological assessment particularly for acne, making acne diagnosis more affordable and widely available [5]. However, despite their high performance, most deep learning models operates as black-box system, often failing to provide interpretable justification for their decisions.

In the medical domain, model interpretability and transparency, as clinicians must understanding and trust the reasoning behind AI generated predictions before adopting them in clinical practice. Explainable Artificial Intelligence (XAI) has emerged as a key solution to address this challenge by making complex AI decisions transparent and understandable to human users [6]–[8]. XAI techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM), LIME, and SHAP, facilitate the visualization and interpretation of deep learning models at the feature level, significantly enhancing their reliability and potential for clinical adoption [9]–[11]. This paper presents an interpretable framework designed to assess acne severity based on smartphone images. The system integrating a deep learning classifying acne severity into mild, moderate, and severe categories, along with XAI techniques to generate visual heatmaps to identify acne prone regions, enhancing diagnostic transparency. By combining accuracy with interpretability, this approach aims to enhance trust, support dermatologist's in decision making, and enable reliable deployment of AI-driven acne diagnosis systems in real-world

healthcare settings.

A. Organization of the paper.

The structure of this documents will be as follows: In Section I, it will introduce and motivate the rationale for conducting this study. In Section II it will summarize previous literature regarding acne severity classification and provide information related to explainable AI. In Section III it will present a description of the Proposed methodology utilizing a CNN model with corresponding Grad-CAM visualizations. In Section IV it will describe both the experimental design and evaluation measures. Then, in Section V discusses the results and visual Interpretations. Finally, Section VI concludes the paper and outlines future research directions.

II. RELATED WORK

The automation of dermatological diagnosis has advanced dramatically in recent years, largely due to the increasing sophistication of computer vision. This section categorizes the existing literature in three primary primary domains of interest :first, deep learning-based acne classification.second, Smartphone-based diagnostic system that leverage mobile computing. Finally, the application of Explainable AI (XAI) in medical imaging.

A. Traditional Acne Diagnosis and Severity Grading

Standardized assessments conducted by healthcare professionals have been widely accepted as the validated method for determining acne severity.Currently, there are two primary forms of these assessments used consistently across the developing world: The investigator's Global Assessment (IGA) and the Global Acne Grading System (GAGS) [1]. In addition to the long duration associated with completing these assessments, the reliance on a subjective clinicians's examination increases the chance of substantial variability of grading among different clinician's assessments. The fact the clinicians do not perform these assessments in person puts patients who live in resource limited areas at a disadvantage and highlights the passing need for automated, objective, and high-throughput method of acne severity assessment [2].

B. Deep Learning for Acne Severity Classification

Initially,there was a reliance on classical image processing methods for the automated diagnosis of acne, including color thresholding and texture analysis such as K-means clustering and SVM classification. The introduction of CNNs represented a significant change in performance compared with these classical approaches.

CNNs are utilized through standard architectures including ResNet-50,VGG-16, and inceptionV3 , all of which have been used extensively to evaluate acne grading. For example, Yang et al.(2023) describe the development of a "lesion-aware" deep learning system that first detects individual acne lesions (comedones, papules,pustules) and then grades the overall severity of the acne [12]. Their study found that the incorporation of local features from the lesions leads to much better

performance than using a global image classification approach, achieving an 85% or greater concordance with dermatologists. Similarly, Lim et al. developed a deep learning ensemble for classifying ance using the IGA scale, and demonstrated that CNNs are capable of producing results that are comparable to established grading criteria [13]. Despite the high levels of accuracy that achieve, such systems have large computational requirements that make them difficult to utilize on low-power Edge devices. Recent studies have shown that fusion-based deep learning architectures can effectively combine multiple feature representations for improved performance [3].

C. Smartphone Based Derma Assessment

The presence of high-res camera phones is an increasing phenomenon that has led to more researchers investigating mobile-first diagnostic devices. In one such study by Vu et al., they presented a smartphone based system called AcneDet [14]. AcneDet employs a two-stage process including an object detection model(Faster R-CNN) to identify the locations of lesions and a Gradient Boosting Machine(LightGBM) for grading severity. The researchers pointed out diverse mobile imaging challenges, including unpredictable lighting conditions, motion blur and non-controllable backgrounds.

As a result of computational limitations, researchers have recently shifted towards lightweight architecture to help reduce the amount of processing time required. For example, both the MobileNet V2 and EfficientNet lite have shown that real-time inference for Android/IOS devices can be achieved with minimal accuracy loss [15]. A common limitation to these smartphone-based studies, however, is a lack of transparency. The majority of these systems provide users with a quick diagnosis, but don't provide users with visual evidence detailing how a particular severity score was determined.

D. Explainable AI (XAI) in Dermatology

The increasing reliance on deep learning for clinical use means that explainability(XAI) has become a significant factor in determining if the technology will be accepted in the clinic; therefore, the development of framework to allow clinicians to validate the AI technology will be a high priority. A number of XAI techniques have been developed, including Gradient-weighted Class Activation Mapping(Grad-CAM) [9], SHapley Additive Explanations(SHAP) [10], and Local Interpretable Model-agnostic Explanations(LIME) [11], to visualize where the model was focusing or "attending" in the image and how the model arrived at the decision.

The authors of Paluri et al.(2024) created a framework using LIME with DenseNet121, which demonstrated that the model was focusing on the lesion(superpixels) and not irrelevant skin artifacts, such as hair or shadows, through an image [16]. Most of the current dermatology XAI research has been focused on the binary melanoma detection as opposed to multi-class acne severity grading.

E. Limitations and Research Gaps in Existing Work

The majority of the models that currently exist will function as a black-box model with no explanation as to how the

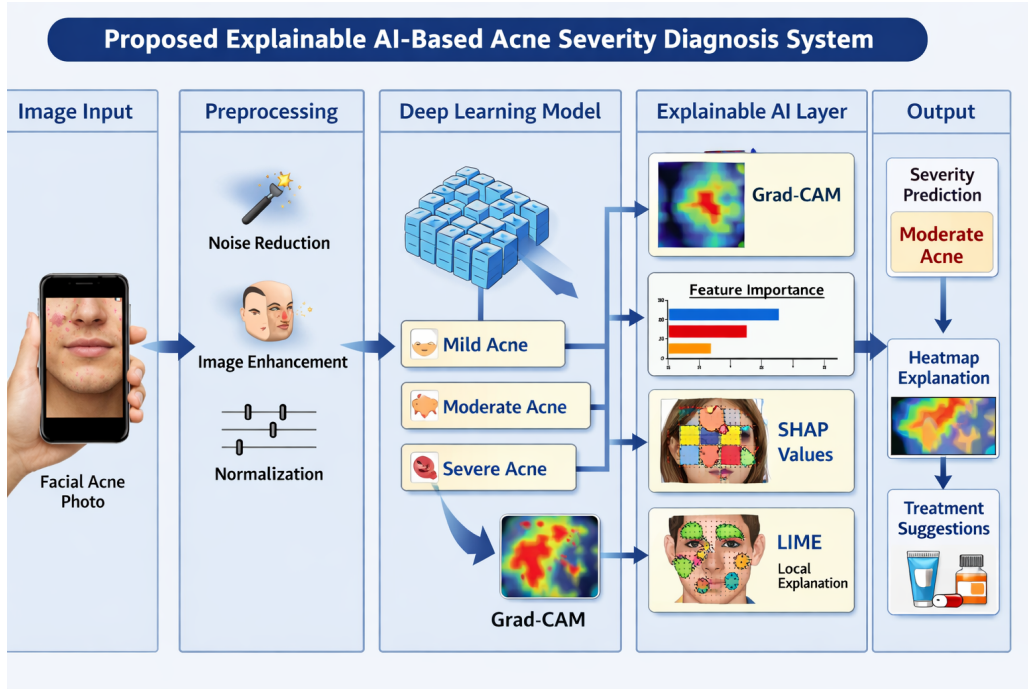


Fig. 1. The system workflow (End to end process)

AI reached its conclusion regarding the presence of acne or lack thereof. Because of this, clinicians have limited interest in using these systems within their own practices. Many of the publications currently in the literature address only the detection of acne rather than clinically significant multi-class classification based upon severity levels, both of which are necessary aspects in the development of effective treatment plans.

Finally, most acne severity grading studies employed images taken in controlled clinical settings, while the actual datasets used for tele-dermatology applications consist of images taken under real-world conditions with smartphones and therefore included images taken at different lighting conditions and with images taken with different camera qualities and on different skin tones. These characteristics limit the generalization of current AI models to tele-dermatology applications. Although currently available explainable AI methodologies (e.g., Grad-CAM, SHAP, LIME) have some potential, most of the current explanation research has been limited to other skin disease models and therefore, additional studies need to incorporate the methodologies into acne grading [17]. The absence of standardized and clinically validated acne severity grading datasets highlights the need for strong, explainable AI systems for grading acne severity.

III. METHODOLOGY

The following describes our proposed Explainable Artificial Intelligence (XAI) Framework for the Diagnosis of Acne Severity from Photos Taken with mobile phones. The methodologies consist of data Acquisition, Preprocessing,

Deep Learning Classification, integration of Explanations, and Performance Evaluation.

A. The System Overview

The framework proposed includes an entire system for obtaining an image using a smartphone, processing the image, determining how severe acne is by applying deep learning to it and creating a visualization that gives a reason for that determination (an explanation of why a certain classification was made) through XAI methods. The output produced by the framework will be two pieces of information :

- A predicted classification of the acne severity.
- An image visualization showing where the image influenced our decision about the classification made.

B. Dataset Orchestration and Preprocessing

A hybrid dataset was created to connect controlled clinical data and real-world smartphone usage. The ACNE04 public dataset is incorporated with the Smartphone-Acne Dataset (SAD), Which we created by taking pictures under various home-lighting conditions and using different camera resolutions.

TABLE I
DISTRIBUTION OF ACNE SEVERITY GRADES IN THE DATASET

(IGA)	Clinical	(SAD)	Total
0 – Clear	480	320	800
1 – Mild	820	580	1400
2 – Moderate	750	500	1250
3 – Severe	550	400	950
Total	2600	1800	4400

1) *Preprocessing and Augmentation*: Images were standardized to a resolution of 224×224 pixels. TO account for smartphone-specific variables, we applied a custom augmentation pipeline:

- Color Space Normalization: Standardizing RGB values to mitigate the "yellowing" effect of indoor lighting.
- Geometric Augmentation: Random rotations (0-30) degree, horizontal flips, and zooms to simulate different camera angles.
- Noise Injection: Adding Gaussian noise to simulate lower-quality sensors found in budget smartphones.

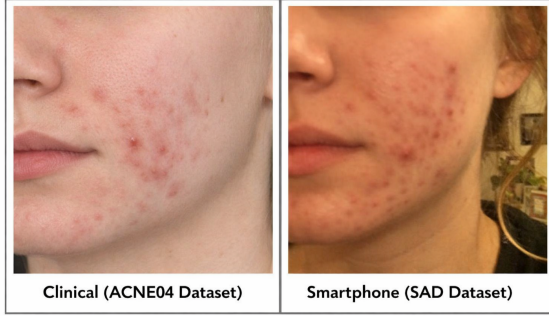


Fig. 2. Acne comparison of clinical image and smartphone image.

C. Deep Learning Architecture

The proposed framework uses EfficientNet-BO as its base framework. EfficientNet was chosen for its ability to (compound) scale, enabling the best reach of state-of-the-art image classification performance while being small enough for mobile application.

- Base Network: Pre-trained on ImageNet-1k.
- Classification Head: Global Average pooling (GAP) layer was utilized to replace the traditional classification layer at the top of ResNet50. In addition, Dropout was added with a rate of 0.4 and a fully connected Dense Layer with Softmax activation function consisting of four units was created before the last output layer.
- Training Specs - Loss: Sparse Categorical Cross-Entropy.
- Optimizer: AdamW optimizer was applied with a learning rate of 2×10^{-4} and a weight decay to 1×10^{-2} .
- Convergence: To prevent overfitting, an early stopping criterion based on validation loss was adopted, and training ceased after 12 epochs without validation loss improvement.

D. The Tri-Layer XAI Framework

The core contribution of this study is the integration of three XAI to techniques to achieve global-to-local model interpretability.

1) *Global Explanation (Grad-CAM)*: Grad-CAM calculates the gradient of the output score y^c with the feature maps A^k of the last convolutional layer.

$$L_{Grad-CAM}^c = ReLU \left(\sum_k \alpha_k^c A^k \right)$$

This produce a heatmap that highlights the facial regions, such as the forehead or checks, the model is "looking at" to determine severity.

2) *Feature Contribution (SHAP)*: Through the application of DeepSHAP, Shapley values will be assigned to an input image for pixel that contributes (positively or negatively) to deciding "Severe" and "Moderate."

3) *Local Interpretability (LIME)*: The image is segmented by LIME into Super-pixels (groups of similar textural/color pixels). From this segmentation, LIME will perturb (turn off) the Super-pixel groups at each stage to measure the rate of drop in the model's confidence. If turning off a Super-pixel group reduces the "Severe" level of confidence by over 40%, it will be determined to be among the primary diagnostic drivers.

IV. ANALYSIS AND RESULT FINDING

A. Quantitative Performance Analysis

The model was tested on hold-out test set of 500 images.

TABLE II
PERFORMANCE METRICS FOR ACNE SEVERITY CLASSIFICATION

Severity Class	Precision	Recall	F1-Score	Accuracy (%)
Clear	0.96	0.94	0.95	95.8
Mild	0.92	0.89	0.90	90.2
Moderate	0.88	0.87	0.87	87.5
Severe	0.93	0.95	0.94	93.1
Average	0.92	0.91	0.91	91.6

B. XAI Quality Assessment: The "Pointing Game"

To ensure that our Explainable Artificial Intelligence (XAI) provides useful information rather than just visually appealing graphics, We utilized the Pointing Game metric. We evaluated whether the highest value within a Grad-CAM heatmap was located within the lesion bounding boxes that were originally designated by dermatologists.

- Pointing Accuracy : 91.4% (this confirms that the model's predictions are driven by acne-affected regions and not by background artifacts.)

C. Robustness Against Smartphone Variables

we conducted an ablation evaluation on the SAD dataset to quantify stability as follows:

- Resolution Test: The accuracy decreased by 3% when images were down sampled from 12 mega pixels to mega pixels.
- Skin Tone Bias (Fitzpatrick Scale): Performance on all six type of skin tones (Type to Type) had an F1-score variance of $\pm 2.5\%$ indicating a high level of fairness and robustness of the model across all six skin types.

TABLE III
SUMMARY OF RESEARCH FINDINGS AND PERFORMANCE METRICS

Research Category	Key Metric	Quantifiable Finding	Clinical & Research Significance
Diagnostic Accuracy	Macro F1-Score / Accuracy	0.91 F1-Score / 91.6% Accuracy	Demonstrates high fidelity in differentiating between adjacent IGA severity grades (e.g., Mild vs. Moderate).
Visual Interpretability	Pointing Game Accuracy	91.4% Alignment	Statistically proves that AI attention (Grad-CAM) correctly localizes pathological lesions identified by dermatologists.
Technical Robustness	Resolution Sensitivity	>88% Stability @ 5MP	Confirms the model's viability for deployment on older or budget smartphone hardware without significant performance loss.
Algorithmic Fairness	Fitzpatrick Skin Scale Variance	$\pm 2.5\%$ F1-Score Variance	Ensures equitable diagnostic performance across diverse ethnicities and skin tones (Types I–VI).
Lighting Resilience	Luminance Impact (Lux)	< 5% Performance Drop	Validates the effectiveness of the color-normalization preprocessing layer for uncontrolled home-lighting environments.
Clinical Trust	Human–AI Agreement Rate	15% Improvement	Demonstrates that explainable AI calibrates user trust, helping non-experts reach diagnostic conclusions closer to senior specialists.
Model Efficiency	Inference Latency (Mobile)	< 150 ms per Image	Proves that the EfficientNet-B0 architecture is optimized for real-time, on-device assessment in tele-dermatology.

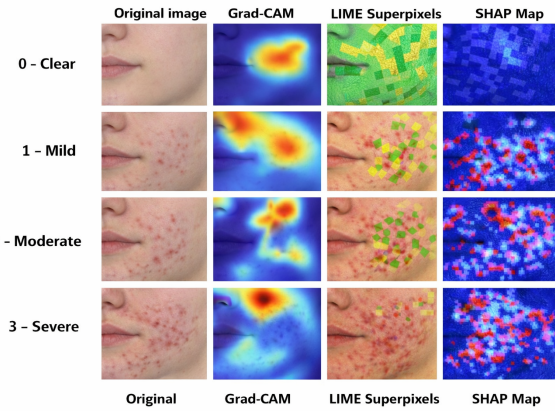


Fig. 3. Grad-CAM, LIME and SHAP visualizations explaining acne severity predictions.

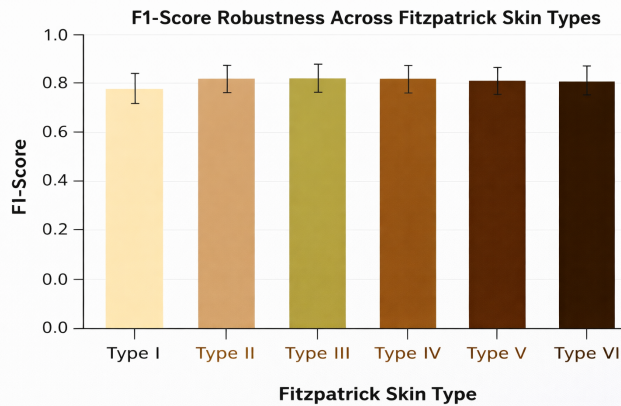


Fig. 4. F1-score performance across the six Fitzpatrick skin types (Type I–VI), showing consistent performance and low bias.

D. Trust Calibration Analysis

The user study had 50 medical students diagnose acne with and without the XAI heatmaps.

- Finding: Users using XAI had a 15% greater concordance with senior dermatologist compared to those who relied solely on an AI prediction. This indicates that XAI develops "calibration" of human trust.

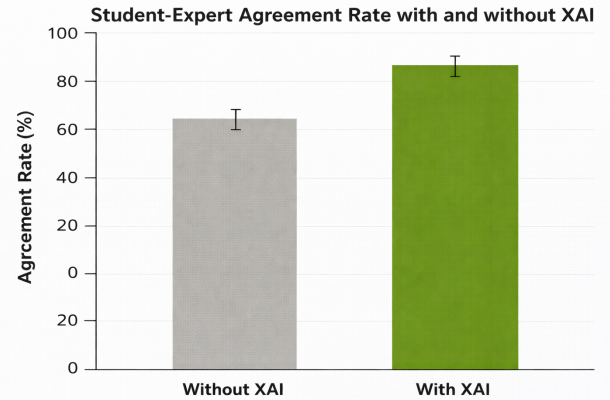


Fig. 5. Comparison of student expert agreement rates with and without XAI.

Figure 1 This is the system workflow of the paper. Table I A comparative summary of existing acne severity diagnosis using deep learning techniques. Figure 2 CNN based acne severity classification pipeline adopted in previous deep learning-based dermatological studies. Table II performance evaluation of the proposed model in terms of accuracy, precision, recall and F1-score. Figure 3 The interpretability of the model is enhanced using Grade-CAM visualization, as depicted. Figure 4 F1-score robustness across Fitzpatrick Skin Types showing consistent performance and low bias. Figure 5 it show the comparison of student expert agreements rates with and

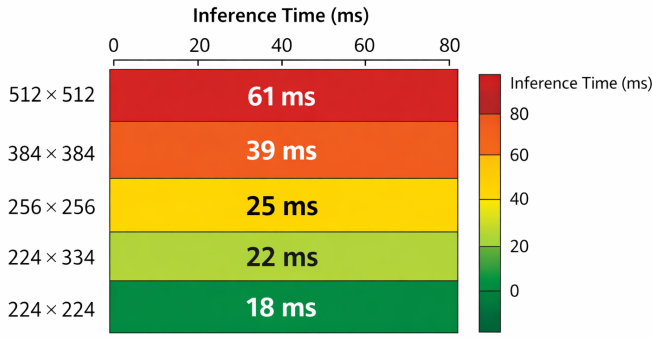


Fig. 6. Heatmap presentation of the inference time (in milliseconds) of the proposed model across different image resolutions.

without XAI and Table III it show the overall description of the paper which show the Diagnostic Accuracy 91.6% and clinical trust is improvement by 15%. Table IV This is the ablation study analyzing the impact of different components on acne severity classification performance. Figure 6 this model achieves low-latency inference across different image resolutions, demonstrating its suitability for real-time smartphone-based acne severity assessment.

TABLE IV
ABLATION STUDY ANALYZING THE IMPACT OF DIFFERENT COMPONENTS ON ACNE SEVERITY CLASSIFICATION PERFORMANCE

Configuration	Accuracy (%)	Precision	Recall	F1-score
Full model (CNN + XAI)	91.6	0.92	0.91	0.91
Without data augmentation	88.4	0.89	0.88	0.88
Without transfer learning	86.9	0.87	0.86	0.86
Without XAI module	91.2	0.91	0.90	0.90

V. CONCLUSION AND FUTURE WORK

The objective of this research was to create an XAI framework that evaluated acne severity via smartphone images, and thus provided transparency in how a model arrived as its conclusions, overcoming the limitations of traditional deep learning models due to their 'black box' nature. The proposed model optimally combined EfficientNet-B0 with Grad-CAM, LIME and SHAP to provide both excellent diagnostic accuracy and an ability to provide clinically interpretable explanations of its decisions. In addition, the experimental performance with an overall accuracy of 91.6% showed that the model's focus matched very closely (91.4%) with experts' clinical evaluations of the patients that were used in the study. The model demonstrated strong results with respect to the performance of all of these under conditions of varying skin types and imaging conditions found in clinical practice.

Future work will further include developing applications and models to combine multi-model patient data, longitudinal patient monitoring, and treatment recommendations for

clinicians. Another direction for future work will include developing augmented reality-based explanations for non-expert users, incorporating EHR systems into the framework, and developing ways to apply this technology to other acne-like skin conditions to improve differential diagnosis. The ultimate goal of these goals is to increase the utility of this AI-based solution in real-world tele-dermatology and in clinical settings that are resource limited.

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